

Report for Cobblestone Gifts Sales Data (Dec 2010-Dec 2011)

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The total revenue for the year was £10,247,282.37. The peak trading season occurs at year end, specifically November, likely a cause of holiday purchases. Below is a table and graph depicting the monthly revenue sales.

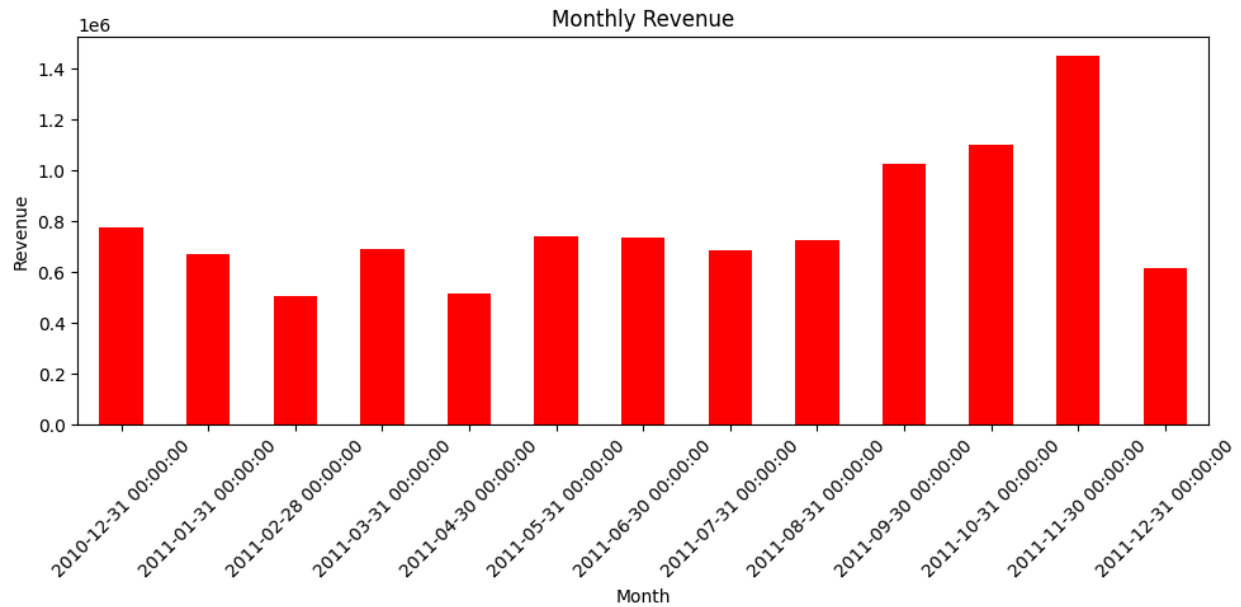


Figure 1: Monthly Revenue (1e6)

Month	Revenue
Dec 2010	775,638.36
Jan 2011	670,380.24
Feb 2011	507,799.87
Mar 2011	689,833.51
Apr 2011	515,407.99
May 2011	739,953.00
Jun 2011	737,558.98
Jul 2011	688,219.34
Aug 2011	724,216.50
Sep 2011	1,028,320.38
Oct 2011	1,103,347.30
Nov 2011	1,452,115.98
Dec 2011	614,490.92

The maximum revenue, in November 2011, is £944,316.11 larger than the minimum revenue, in February 2011. It is clear from the Figure that peak revenue comes in the later months of the year.

The top ten best-selling products, with stock codes and descriptions, by their yearly revenues, are listed below.

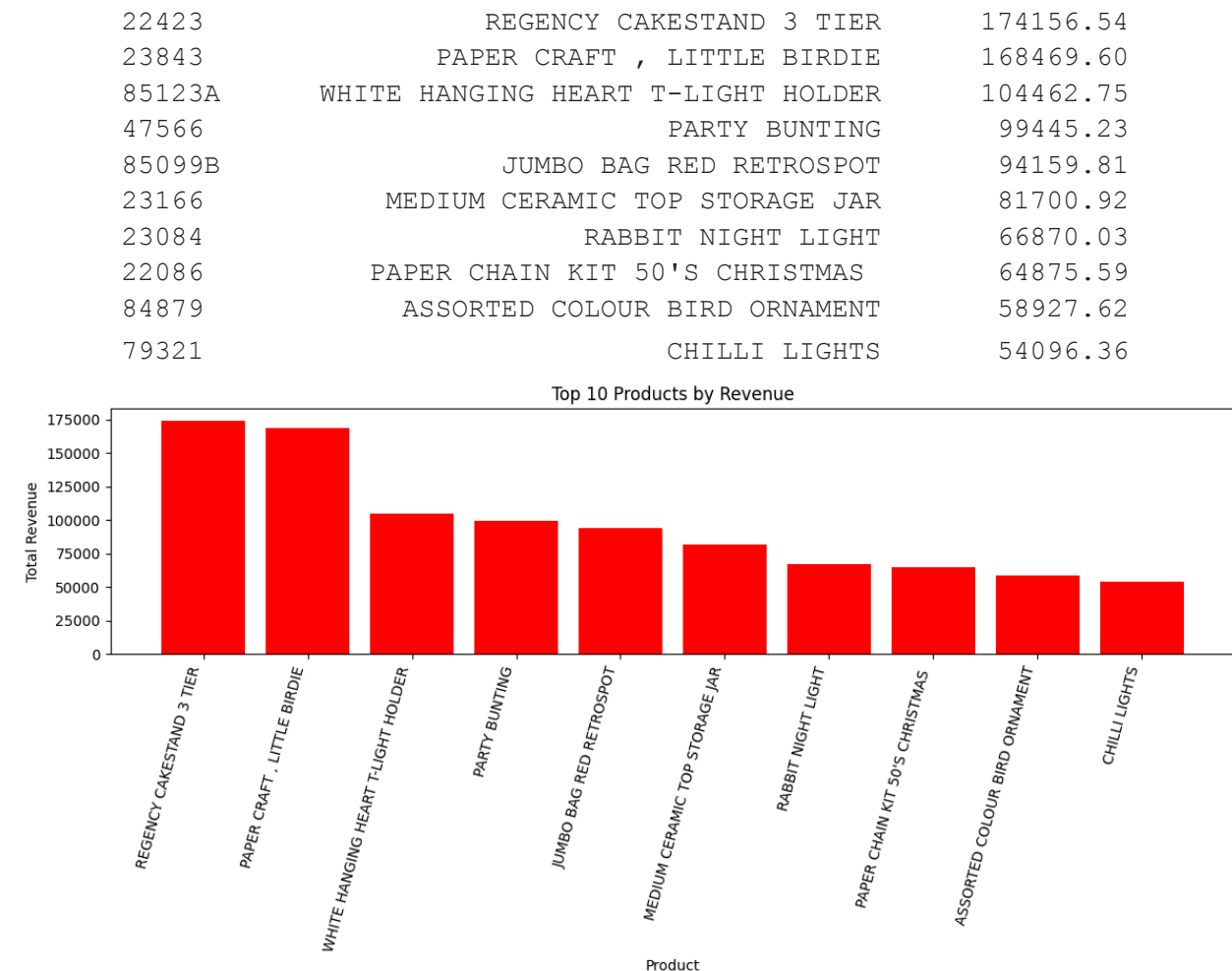


Figure 2: Top Products by Revenue

The top ten best-selling products, with stock codes and descriptions, by their yearly sold quantities, are listed below.

23843	PAPER CRAFT , LITTLE BIRDIE	80995
23166	MEDIUM CERAMIC TOP STORAGE JAR	78033
22197	POPCORN HOLDER	56898
84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS	54951
85099B	JUMBO BAG RED RETROSPOT	48371
85123A	WHITE HANGING HEART T-LIGHT HOLDER	37641
21212	PACK OF 72 RETROSPOT CAKE CASES	36396
84879	ASSORTED COLOUR BIRD ORNAMENT	36362
23084	RABBIT NIGHT LIGHT	30739
22492	MINI PAINT SET VINTAGE	26633

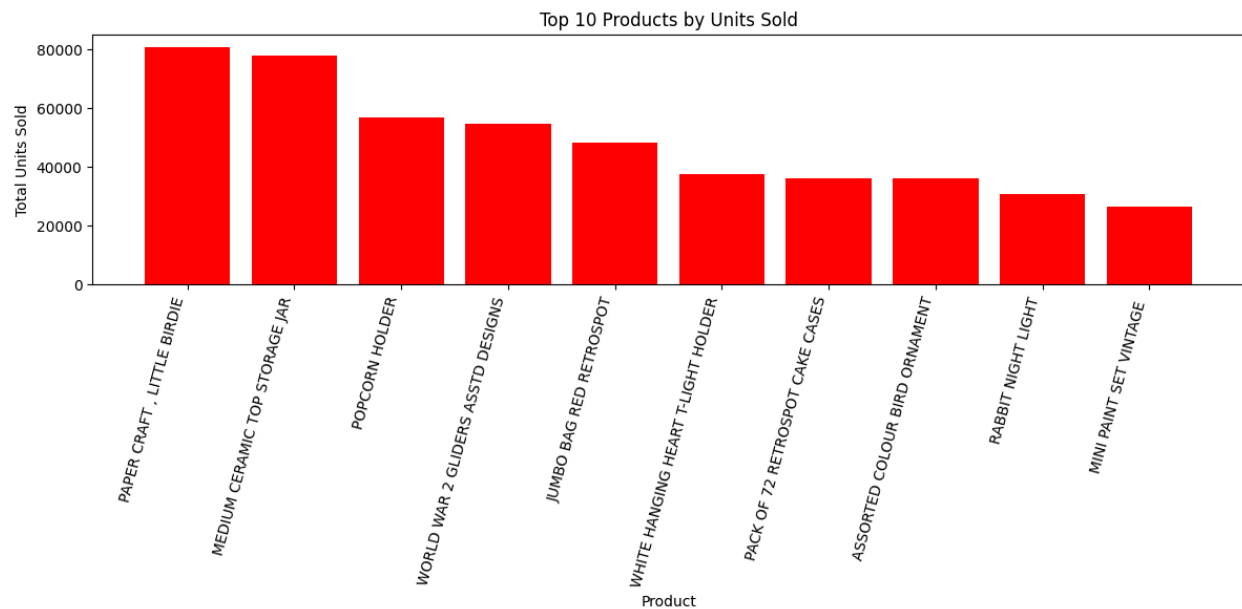


Figure 3: Top Products by Quantity Sold

There are six products that are included in both lists, and the following lists detail items that are only in *one* of the above categories.

Only in top10 revenue: ['CHILLI LIGHTS', 'REGENCY CAKESTAND 3 TIER', 'PAPER CHAIN KIT 50'S CHRISTMAS', 'PARTY BUNTING']

These four items are likely high priced, meaning they do not need a high quantity of sales to rack up a significant amount of revenue.

Only in top10 units: ['MINI PAINT SET VINTAGE ', 'POPCORN HOLDER', 'WORLD WAR 2 GLIDERS ASSTD DESIGNS', 'PACK OF 72 RETROSPOT CAKE CASES']

These four items are likely low priced, meaning their high quantity of sales do not contribute to a significant amount of revenue. They could likely be increased in price due to their high demand.

The table below gives the top 10 countries by revenue, excluding the United Kingdom.

	total_revenue	distinct_customers	revenue_per_customer
country			
Netherlands	283889.34	9	31543.26
Ireland	270850.86	4	67712.72
Germany	205381.15	94	2184.91
France	184493.00	88	2096.51
Australia	138103.81	9	15344.87
Spain	55706.56	30	1856.89
Switzerland	53065.60	22	2412.07
Japan	37416.37	8	4677.05
Belgium	36927.34	25	1477.09
Sweden	36828.83	8	4603.60

The table below gives the top 10 countries by customer count, excluding the United Kingdom.

	total_revenue	distinct_customers	revenue_per_customer
country			
Germany	205381.15	94	2184.91
France	184493.00	88	2096.51
Spain	55706.56	30	1856.89
Belgium	36927.34	25	1477.09
Switzerland	53065.60	22	2412.07
Portugal	26951.11	20	1347.56
Italy	15820.24	14	1130.02
Finland	18344.88	12	1528.74
Austria	8742.68	11	794.79
Norway	32454.64	10	3245.46

The table below gives the top 10 regions by revenue, excluding the United Kingdom in the UK & Ireland region.

	total_revenue	distinct_customers	revenue_per_customer
region			
Western Europe	728354.15	235	3099.38
UK & Ireland	290848.40	13	22372.95
Asia/Oceania	194384.76	19	10230.78
Southern Europe	158039.62	91	1736.70

Northern Europe	111810.75	41	2727.09
Middle Eastern	25755.29	18	1430.85
North America	6695.83	8	836.98
Missing	4740.94	5	948.19
South America	1143.60	1	1143.60
Africa	1002.31	1	1002.31

There is a notable domination in Western Europe, but the revenue per customer lacks in this region. The domination stems from the amount of distinct customers. Based on proximity, the lack of revenue from the Northern Europe region is surprisingly significantly less than other European regions, and Oceania.

The past year, the top ten spenders contributed to 31.99% of the revenue. These wholesale buyers are dominating the revenue income, spending £3,278,105.63. Their identifications and corresponding revenue sums are listed below.

14646.0	279138.02
18102.0	259657.30
17450.0	194390.79
16446.0	168472.50
14911.0	136161.83
12415.0	124564.53
14156.0	116560.08
17511.0	91062.38
12346.0	77183.60
16029.0	72708.09

Returning to the comparison between the United Kingdom and non-United Kingdom orders, the average order value for both markets are expressed below.

	(Avg Order Value)	(Median Order Value)	(# of Orders)
Market: Non-UK	813.45	385.28	1872
Market: UK	487.32	299.60	17903

The Non-UK market typically sees larger invoices. This is likely due to bulk-purchasing, which would most likely come from international purchases. This would reduce shipping costs for these customers, purchasing higher quantities of items at one time would increase their invoice totals. The UK dominance in the sales data is clearly illustrated in this data presentation as well.

There are a significant amount of cancellations in the data presented, 1.72% of the data is a cancellation, making up 8.41% of the revenue data.

The table below shows the top ten cancelled items by their cancelled revenue value. Notably, the single return of the Paper Craft, Little Birdie is a mistake. But, 181 canceled Regency Cakestand 3 Tier is not. This is a very common purchase, top 10 in items sold, which means it will get returned more often.

stock_code	description	cancel_count	cancel_value
23843	PAPER CRAFT , LITTLE BIRDIE	1	-168469.60
23166	MEDIUM CERAMIC TOP STORAGE JAR	10	-77479.64
22423	REGENCY CAKESTAND 3 TIER	181	-9722.55
85123A	WHITE HANGING HEART T-LIGHT HOLDER	42	-6624.30
21108	FAIRY CAKE FLANNEL ASSORTED COLOUR	3	-6591.42
23113	PANTRY CHOPPING BOARD	6	-4803.06
48185	DOORMAT FAIRY CAKE	3	-4554.90
21175	GIN + TONIC DIET METAL SIGN	7	-3775.33
47566B	TEA TIME PARTY BUNTING	7	-3692.95
22273	FELTCRAFT DOLL MOLLY	6	-3512.65

The table below shows the top ten cancelled customer identifications, by their cancelled revenue value. Excluding NO_ID, a filler for missing customer data, most single count cancellations can be excluded. But, customer 14911 has 226 cancelled items to their ID.

customer_id	cancel_count	cancel_value
NO_ID	383	-285470.40
16446.0	1	-168469.60
12346.0	1	-77183.60
15098.0	2	-39267.00
16029.0	32	-30032.23
15749.0	5	-22998.40
12744.0	7	-12158.90
14911.0	226	-11252.44
12931.0	20	-8593.15
12536.0	12	-8495.01

The tables below show the top ten cancellations by their cancelled amounts, both by description and by customer. These tables highlight the data points mentioned above.

stock_code	description	cancel_count
22423	REGENCY CAKESTAND 3 TIER	181
22960	JAM MAKING SET WITH JARS	87
22720	SET OF 3 CAKE TINS PANTRY DESIGN	74
21232	STRAWBERRY CERAMIC TRINKET BOX	61

S	SAMPLES	61
22699	ROSES REGENCY TEACUP AND SAUCER	54
22197	POPCORN HOLDER	50
22666	RECIPE BOX PANTRY YELLOW DESIGN	47
82483	WOOD 2 DRAWER CABINET WHITE FINISH	45
20725	LUNCH BAG RED RETROSPOT	44

	cancel_count	cancel_value
customer_id		
NO_ID	383	-285470.40
14911.0	226	-11252.44
17841.0	136	-650.79
17511.0	113	-2937.00
15311.0	112	-1348.56
12607.0	101	-1579.51
14410.0	93	-528.77
13798.0	90	-802.43
14606.0	82	-442.80
13113.0	79	-1735.96

Finally, some data was altered for this report. The number of rows that are not “typical” are given, with their reason, in the table below.

reason	
cancellation	9288
duplicate	5221
non_product_code	2356
bad_price	577
bad_quantity	474

Cancellations were excluded during all sales analysis. They do not exist in the cleaned data. Exact row duplicates were removed and assumed to be data errors, not exact repeat purchases. Non-product codes were removed from sales analysis. Bad price and bad quantity rows that were not included in cancellations were assumed to be errors as well, requiring their elimination. Any missing data for customer_ID was included in the analysis, because it does not affect any of the sales metrics in question. In total, 17,196 rows were disregarded in clean sales analysis, which is only 3.17% of the data. Therefore, we can use the data for this analysis with reasonable accuracy.